

# Electricity: Getting the timing right

**P**rices in a perfect market are determined by demand and supply. Both demand and supply change, and so do prices. Consumers are often exposed to these changing dynamics. Think of how airline tickets jump before the vacation travel season. Or how one has to pay more for watching a movie on weekends compared to weekdays. Now, imagine the same scenario for electricity. Experts call it dynamic retail pricing, or “Time-of-Day” (ToD) tariff. ToD tariff is recognised globally as an important demand side management tool for incentivising consumers to shift a portion of their loads from peak times to off-peak times, thereby improving the system load factor.

Many countries across the world have implemented ToD tariffs. These include the US, Canada, UK, Australia, Japan, Germany, France, Italy, Spain and Sweden. Japan has even introduced a seasonal variation called Cool Biz. In summer months, this programme encourages businesses to reduce their electricity usage during peak hours by adjusting air-conditioning settings.

The Government of India has recently passed an amendment to the Electricity (Rights of Consumers) Rules, 2020, making two important changes. These relate to the introduction of the ToD tariff system and rationalisation of smart metering provisions. The ToD tariff would be applicable for commercial and industrial consumers having a maximum demand of 10 KW and above from April 1, 2024; and for all other consumers, except agricultural consumers, from April 1, 2025. Tariffs during “solar hours” (duration of eight hours in a day), as specified by the state electricity regulatory commissions, shall be 10 to 20 per cent less than the normal tariff, while the tariff during peak hours will be 10 to 20 per cent higher than normal. Many state electricity regulatory commissions have already imposed a ToD regime for large commercial and industrial categories.

ToD tariff is to be made effective immediately after

the installation of smart meters. Hence, their success, especially at the consumer level, is contingent on smart metering being aggressively implemented, and users being made aware of how to optimise their consumption patterns to take advantage of the scheme. Smart meters send consumption information to power distribution companies every 15 minutes, which is crucial for calculating ToD charges. Currently, over 6.5 million smart meters have been installed in the country, with a target of reaching 250 million by 2026, under the government’s Revamped Distribution Sector Scheme (RDSS). The government launched the RDSS with an outlay of ₹3.03 trillion for five years from FY21-22 to FY25-26. A large portion is earmarked for smart metering.



## INFRATALK

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ToD tariff is happening at a time in India when energy transition to renewables is no longer a buzzword. By the middle of this year, renewable capacity will have reached 127 GW. From FY15 to FY23, solar and wind power generation has grown almost three times, from 5.5 per cent to 14.3 per cent. The good news is that this helps India in its climate goals. However, the bad news is that the wind does not blow all the time and the sun sets at the end of each day. Storage can help, but at this time, there are technical and commercial challenges to mass-scale energy storage. This thus makes the proposed introduction of ToD pricing timely and relevant.

One of the basic tenets of grid operations is that electricity supply has to match demand at all times. Any imbalance leads to system failures, which can have serious consequences. But weather and time dependent resources, like wind and solar power, cannot perform according to the grid operator’s needs. This is where the demand side of the equation comes in handy. Instead of tweaking the supply side, if price-based signals can be provided to consumers, they can then be motivated to alter their demand in line with system requirements, and ease

potential disruptions to the grid.

ToD is clearly expected to lead to changes in consumer behaviour vis-à-vis shifting usage to off-peak hours for applications such as electric vehicle charging, washing machine, and cooling. In India, most consumers are aware that the household appliance that consumes maximum electricity is the air-conditioner. There could possibly be some negatives too for consumers. These could relate to:

- Increased complexity: Time of day pricing introduces more complexity into the billing system. Consumers may need to track and manage their electricity usage during different time periods, which can be challenging and time-consuming.
- Behavioural changes: This may require changes in consumer behaviour, such as running appliances or charging electric vehicles at specific times. Not all consumers may be willing or be able to make these adjustments.
- Increased costs during peak hours: If consumers are unable to shift their usage to off-peak times, they may end up paying higher prices during peak hours.

Commenting on the development, Union Power Minister R K Singh said: “The ToD tariff will improve the management of renewable generation fluctuations, incentivise demand increase during the periods of high RE generation hours and thereby increase grid integration of larger quantities of renewable power.” The move is also expected to help India work towards its target of achieving 65 per cent of its energy capacity from non-fossil fuels by 2030 and net zero emissions by 2070.

Above all, the scheme effectively rewards those consumers who are willing to exercise their choice of energy consumption in response to time-varying rates. For too long such consumers have been mute participants in the energy transition movement. They can now be willing partners!

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